

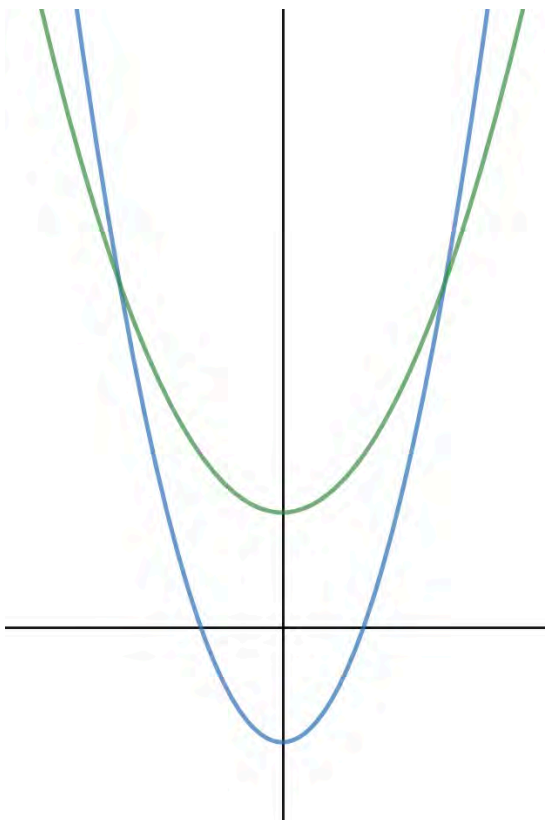
Question 1 (10 marks) Use a SEPARATE writing booklet

a. $\int_{-2}^1 x^3 - x + 3 \, dx$ 2

b. $\int \frac{x^4 + x^2 - 2}{x^2} \, dx$ 2

c. Find the exact area contained by the function $f(x) = \sqrt{3x - 6}$, the x axis and the line $x = 5$. 3

d. i) Find the points of intersection of the curves $y = 2x^2 - 1$ and $y = x^2 + 1$ 1

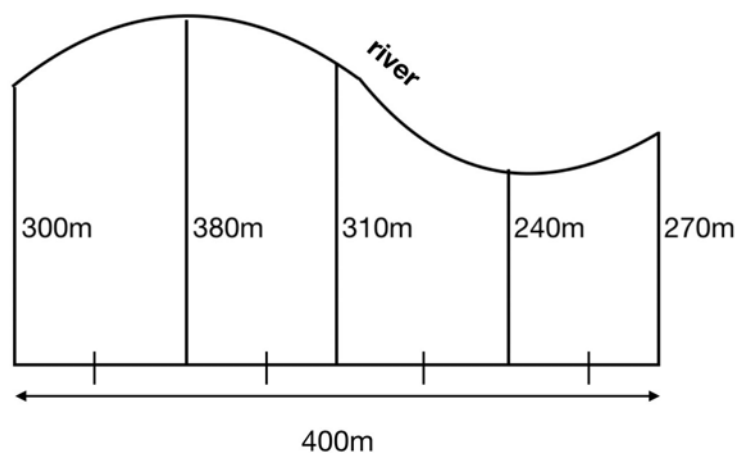


ii) Find the exact area contained between the curves $y = 2x^2 - 1$ and $y = x^2 + 1$ 2

End of Question 1

Question 2 (10 marks) Use a SEPARATE writing booklet

- a. A nature reserve borders a river as seen in the diagram below. An environment and landscape management company needs to know the area of this reserve to quote a price to maintain it. 2

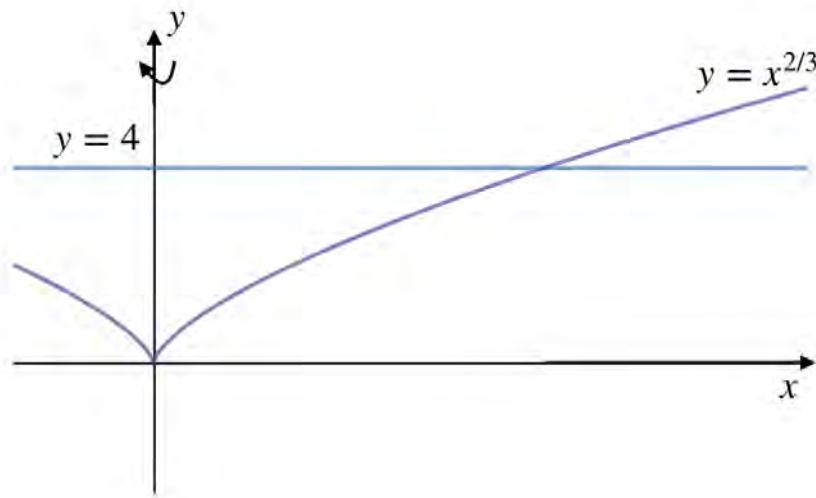


Use the trapezoidal rule to estimate the area of the nature reserve.

- b. i) Differentiate $f(x) = (2x^2 - 3)^8$ 1
- ii) Hence, or otherwise, integrate $g(x) = x(2x^2 - 3)^7$ 2
- c. Use Simpson's rule with 5 function values to evaluate: 2

$$\int_0^2 \frac{dx}{x+2}$$

- d. Find the volume of the solid of revolution formed when the area bounded by the y axis, the curve $y = x^{\frac{2}{3}}$ and the line $y = 4$ is rotated about the y axis. 3



End of Question 2

Question 3 (9 marks) Use a SEPARATE writing booklet

- a. Use the first derivative to find values of x for which $f(x) = x(12 - x^2)$ is increasing. **3**
- b. A function $f(x) = Ax^3 + Bx^2 + Cx + 1$ has a local maximum at point $(-1, 3)$ and a point of inflection at point $(\frac{-1}{3}, \frac{49}{27})$. Find the values of A , B and C . **3**
- c. Find the equation of the normal to the curve $y = \frac{x^2}{3-x^2}$ at the point $(1, \frac{1}{2})$. **3**

End of Question 3

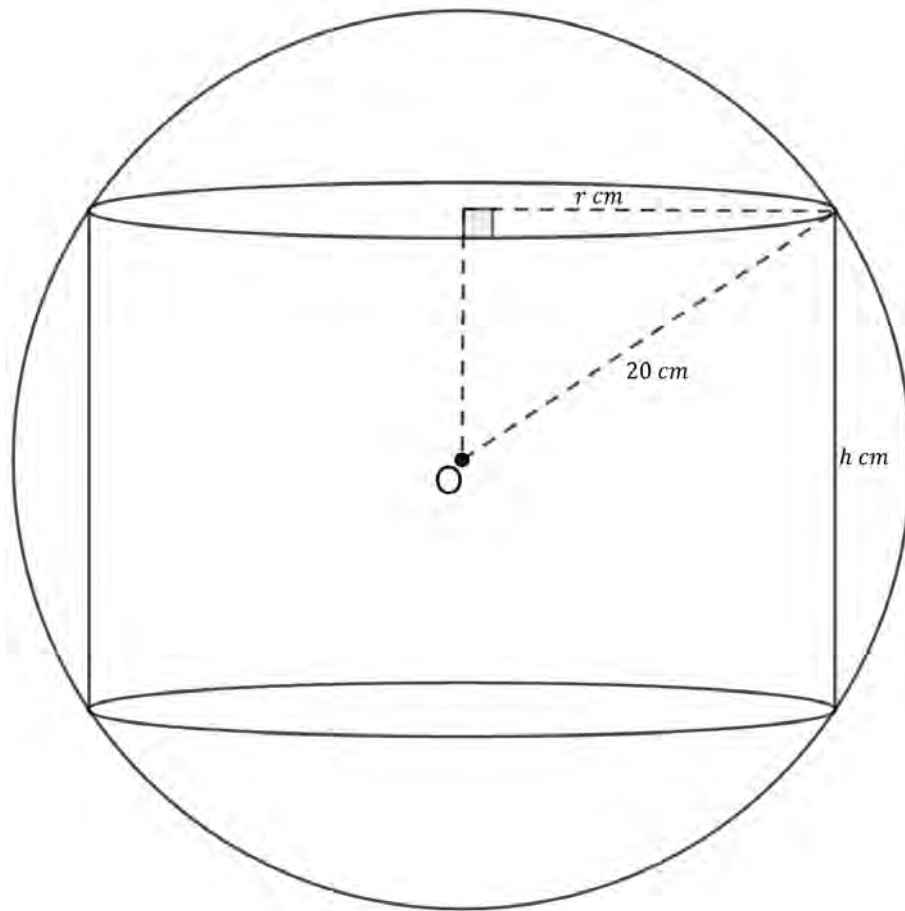
Question 4 (11 marks) Use a SEPARATE writing booklet

a. A function $y = f(x)$ has the following characteristics:

- The graph is a smooth, continuous curve and the function is differentiable for all real x
- $f'(x) = 0$ when $x = 2$ and -7
- $f''(x) < 0$ for $-4 < x < 2$
- $f''(x) > 0$ for $x < -4$ and $x > 2$

- | | |
|--|----------|
| i) Identify the x values of any points of inflection. | 1 |
| ii) Identify the x values of all stationary points and determine their nature. | 2 |
| iii) Sketch a possible graph of the function, clearly showing the x value of each key point. | 2 |

- b. A solid cylinder is to be cut from a sphere of radius 20cm. O is the centre of both the sphere and cylinder.



- i) Find an expression for the radius r of the cylinder in terms of its height h . 1
- ii) Hence or otherwise, show that the volume of the cylinder is given by: 2

$$v = 400\pi h - \frac{\pi h^3}{4}$$
- iii) Find the height that would maximise the volume of the cylinder. 3

End of Examination

Solutions

Tuesday, 26 February 2019

7:22 PM

Question One

a) $\int_{-2}^1 x^3 - x + 3 \, dx$

$$= \left[\frac{x^4}{4} - \frac{x^2}{2} + 3x \right]_{-2}^1$$

| mark for integration

$$= \left(\frac{1^4}{4} - \frac{1^2}{2} + 3 \cdot 1 \right) - \left(\frac{(-2)^4}{4} - \frac{(-2)^2}{2} + 3 \cdot (-2) \right)$$

$$= \left(\frac{1}{4} - \frac{1}{2} + 3 \right) - \left(\frac{16}{4} - \frac{4}{2} - 6 \right)$$

$$= \frac{11}{4} \quad \dots 4$$

$$= \frac{27}{4}$$

| mark for answer

b) $\int \frac{x^4 + x^2 - 2}{x^2} \, dx$

$$= \int x^2 + 1 - \frac{2}{x^2} \, dx$$

| mark for integration

$$= \frac{x^3}{3} + x + \frac{2}{x} + C$$

| mark for constant 'C'

c) $f(x) = 0$ when $\sqrt{3x-6} = 0$
 $3x-6=0$
 $3x=6$
 $x=2$



$$\text{Area} = \int_2^5 \sqrt{3x-6} \, dx$$

| mark for boundaries

$$= \int_2^5 (3x-6)^{\frac{1}{2}} \, dx$$

$$= \left[\frac{(3x-6)^{\frac{3}{2}}}{\frac{3}{2} \times 3} \right]_2^5$$

| mark for integration

$$= \left[\frac{2}{9} (3x-6)^{\frac{3}{2}} \right]_2^5$$

$$= \frac{2}{9} \left[(3 \cdot 5 - 6)^{\frac{3}{2}} - (3 \cdot 2 - 6)^{\frac{3}{2}} \right]$$

$$= \frac{2}{9} \left((3 \cdot 5 - 6)^{\frac{3}{2}} - (3 \cdot 2 - 6)^{\frac{3}{2}} \right)$$

$$= \frac{2}{9} \left(9^{\frac{3}{2}} - 0^{\frac{3}{2}} \right)$$

$$= \frac{2}{9} (9^2 - 0^2)$$

$$= \frac{2}{9} \times 27$$

$$= 6 \text{ units}^2$$

1 mark for answer

$$d) y = 2x^2 - 1 \quad y = x^2 + 1$$

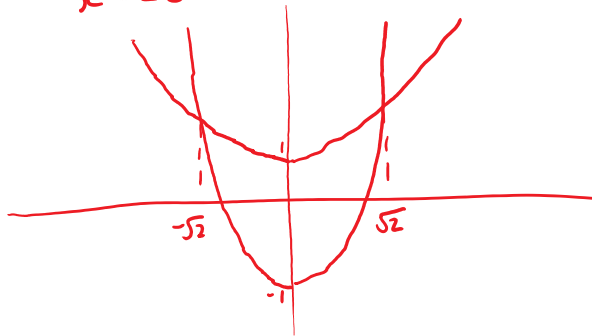
i) For points of intersection

$$2x^2 - 1 = x^2 + 1$$

$$x^2 = 2$$

$$x = \pm\sqrt{2}$$

1 mark



$$ii) A = \int \text{top} - \text{bottom} \, dx$$

$$= \int_{-\sqrt{2}}^{\sqrt{2}} (x^2 + 1) - (2x^2 - 1) \, dx \quad | \text{ mark for correct integral}$$

$$= \int_{-\sqrt{2}}^{\sqrt{2}} -x^2 + 2 \, dx$$

$$= \left[-\frac{x^3}{3} + 2x \right]_{-\sqrt{2}}^{\sqrt{2}}$$

$$= \left(\left(-\frac{\sqrt{2}^3}{3} + 2\sqrt{2} \right) - \left(-\frac{(-\sqrt{2})^3}{3} - 2\sqrt{2} \right) \right)$$

$$= \left(-\frac{2\sqrt{2}}{3} + \frac{6\sqrt{2}}{3} \right) - \left(\frac{2\sqrt{2}}{3} - \frac{6\sqrt{2}}{3} \right)$$

$$= \frac{8\sqrt{2}}{3} \text{ units}^2$$

1 mark for answer

Question Two

$$a) h = \frac{400}{4}$$

1 mark for $h=100$ or $b-a=100$

Question 1

a) $h = \frac{400}{4}$
 $= 100$

| mark for $h=100$ or $b-a=100$

$$A \approx \frac{100}{2} (300 + 2 \times (380 + 310 + 240) + 270)$$

$$\approx 121500 \text{ m}^2$$

| mark for answer

b) $f(x) = (2x^2 - 3)^8$

$$f'(x) = 8 \cdot 4x (2x^2 - 3)^7$$

$$= 32x (2x^2 - 3)^7$$

| mark for derivative

ii $g(x) = x (2x^2 - 3)^7$

$$\int x (2x^2 - 3)^7 dx = \frac{1}{32} \int 32x (2x^2 - 3)^7 dx$$

| mark for taking $\frac{1}{32}$

$$= \frac{1}{32} ((2x^2 - 3)^8 + C)$$

$$= \frac{2x^2 - 3^8}{32} + C$$

| mark for answer

c)

x	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
$f(x)$	$\frac{1}{2}$	$\frac{2}{5}$	$\frac{1}{3}$	$\frac{2}{7}$	$\frac{1}{4}$
Weight	1	4	2	4	1

$$h = \frac{1}{2}$$

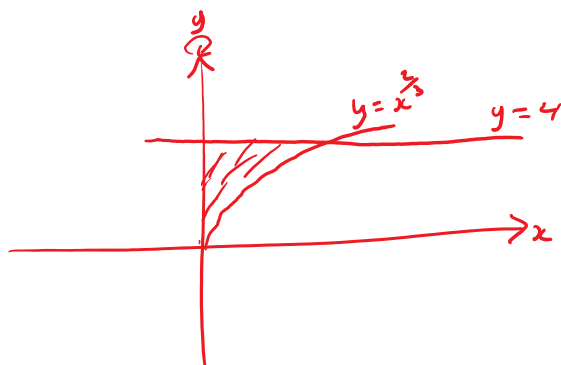
| mark for $h = \frac{1}{2}$ or $b-a = \frac{1}{2}$

$$\int_0^2 \frac{dx}{x+2} \approx \frac{1}{2} \left(\frac{1}{2} + 4 \times \left(\frac{2}{5} + \frac{2}{7} \right) + 2 \times \frac{1}{3} + \frac{1}{4} \right)$$

$$\approx \frac{1747}{2520} \text{ or } 0.693 \text{ to 3 d.p.}$$

| mark for answer

d)



$$V = \pi \int x^2 dy$$

For $y = x^{\frac{2}{3}}$

$$y^3 = x^2$$

| mark for $x^2 = y^3$

| mark for boundaries

$$V = \pi \int_0^4 y^3 dy$$

$$= \pi \left[\frac{y^4}{4} \right]_0^4$$

$$= \pi \left((4^3) - (0) \right)$$

$$= 64\pi \text{ units}^3$$

| mark for answer

Question Three

$$f(x) = x(12 - x^2)$$

$$= 12x - x^3$$

Increasing $\Rightarrow f'(x) > 0$ | mark for $f'(x) > 0$

$$f'(x) = 12 - 3x^2$$
 | mark for derivative

$$12 - 3x^2 > 0$$

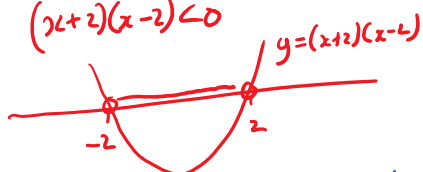
$$12 > 3x^2$$

$$4 > x^2$$

$$x^2 < 4$$

$$x^2 - 4 < 0$$

$$(x+2)(x-2) < 0$$



$$\therefore -2 < x < 2$$

| mark for answer

i.e. $f(x)$ is increasing when $-2 < x < 2$

b) $f(x) = Ax^3 + Bx^2 + Cx + 1$

$$\text{max TP at } (-1, 3) \Rightarrow f'(x) = 0, f''(x) < 0$$

$$\text{Point of inflection at } \left(-\frac{1}{3}, \frac{49}{27}\right) \Rightarrow f''(x) = 0$$

$$\text{Now } f'(x) = 3Ax^2 + 2Bx + C$$

Now $f'(x) = 3Ax^2 + 2Bx + C$

and

$$f''(x) = 6Ax + 2B$$

When $x = -\frac{1}{3}$ $f''(x) = 0$

$$f''(-\frac{1}{3}) = 6A(-\frac{1}{3}) + 2B = 0$$

$$-2A + 2B = 0$$

$$2B = 2A$$

$$\boxed{B = A} \quad (1)$$

When $x = -1$, $f'(x) = 0$

| mark for substituting $x = -1$ in $f'(x) = 0$

$$f'(x) = 3Ax^2 + 2Bx + C \quad \text{sub (1)}$$

$$= 3Ax^2 + 2Ax + C$$

$$f'(-1) = 3A(-1)^2 + 2A(-1) + C = 0$$

$$3A - 2A + C = 0$$

$$A + C = 0$$

$$\boxed{C = -A} \quad (2)$$

Subbing (1) and (2)

$$f(x) = Ax^3 + Ax^2 - Ax + 1$$

$$f(-1) = 3$$

$$= A(-1)^3 + A(-1)^2 - A(-1) + 1$$

$$= -A + A + A + 1$$

$$= A + 1 = 3$$

$$A = 2$$

$$\therefore f(x) = 2x^3 + 2x^2 - 2x + 1 \quad | \text{ mark for answer}$$

c) $y = \frac{x^2}{3-x^2} \quad (1, \frac{1}{2})$

Let $u = x^2 \quad u' = 2x$

$$v = 3-x^2 \quad v' = -2x$$

$$\therefore y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{(3-x^2) \cdot 2x - x^2 \cdot (-2x)}{(3-x^2)^2}$$

$$= \frac{6x - 2x^3 + 2x^3}{(3-x^2)^2}$$

$$\begin{aligned}
 & (3-x^2)^{-1} \\
 &= \frac{6x-2x^3+2x^3}{(3-x^2)^2} \\
 &= \frac{6x}{(3-x^2)^2} \quad | \text{ mark for derivative}
 \end{aligned}$$

$$\begin{aligned}
 \text{When } x=1 \quad y' &= \frac{6 \cdot 1}{(3-1^2)^2} \\
 &= \frac{6}{4} \\
 &= \frac{3}{2}
 \end{aligned}$$

$\therefore$ Gradient of tangent is $\frac{3}{2}$
and

gradient of normal is $-\frac{2}{3}$ | mark for gradient of normal

$$y - \frac{1}{2} = -\frac{2}{3}(x-1)$$

$$= -\frac{2x}{3} + \frac{2}{3}$$

$$y = \frac{7}{6} - \frac{2x}{3} \quad \text{or} \quad 4x + 6y - 7 = 0 \quad | \text{ mark for answer}$$

Question Four

ai) Points of inflection at $x=2, -4$ (as concavity changes) | mark

ii) Stationary points $\Rightarrow f'(x)=0$ at $x=2, -7$

| mark for x-values

Also a point of inflection at $x=2$

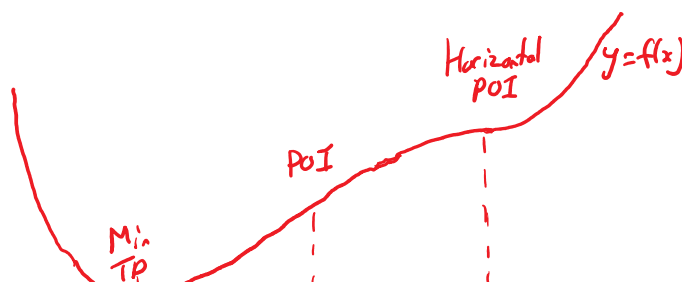
$\therefore$ Horizontal point of inflection at $x=2$

$$f''(-7) > 0$$

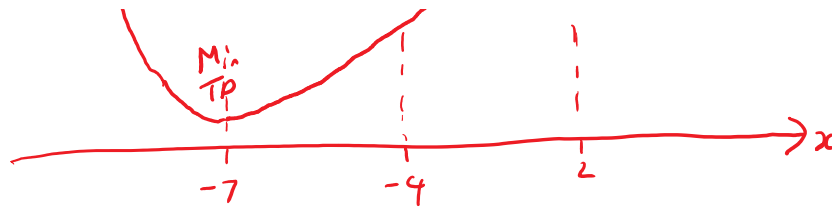
$\therefore$ Minimum turning point at $x=-7$

| mark for nature
Hor POI
Min TP

iii)



- | mark for each incorrect/missing
 * Shape
 * Label of x-values
 * Type of key point



* Label of x-values
* Type of key point

$$\text{b)} \quad r^2 = 20^2 - \left(\frac{h}{2}\right)^2$$

$$= 400 - \frac{h^2}{4}$$

$$r = \sqrt{400 - \frac{h^2}{4}} \quad | \text{ mark for answer}$$

$$\text{ii)} \quad V = \pi r^2 h$$

$$= \pi \left(400 - \frac{h^2}{4}\right) h \quad | \text{ mark for substitution}$$

$$= 400\pi h - \frac{\pi h^3}{4} \quad \text{as required} \quad | \text{ mark for answer}$$

$$\text{iii)} \quad \frac{dV}{dh} = 400\pi - \frac{3\pi h^2}{4} \quad | \text{ mark for derivative}$$

$$\frac{dV}{dh} = 0 \quad \text{when} \quad 400\pi - \frac{3\pi h^2}{4} = 0$$

$$\frac{3\pi h^2}{4} = 400\pi$$

$$h^2 = 400 \times \frac{4}{3}$$

$$= \frac{1600}{3}$$

$$h = \pm \sqrt{\frac{1600}{3}}$$

$$= \pm \frac{40}{\sqrt{3}}$$

$$= \pm \frac{40\sqrt{3}}{3}$$

Ignore the negative case
as $h > 0$

$$h = \frac{40\sqrt{3}}{3} \quad | \text{ mark for height}$$

$$d^2V$$

$$-\frac{\pi h}{4}$$

$$\therefore d^2V < 0 \text{ for any } h > 0$$

$\frac{d^2h^2}{dh^2}$

$\therefore$ Max turning point at $h = \frac{40\sqrt{3}}{3}$

i.e. Maximum volume when height is $\frac{40\sqrt{3}}{3} \text{ m}$

| mark for finding
nature of
stationary point